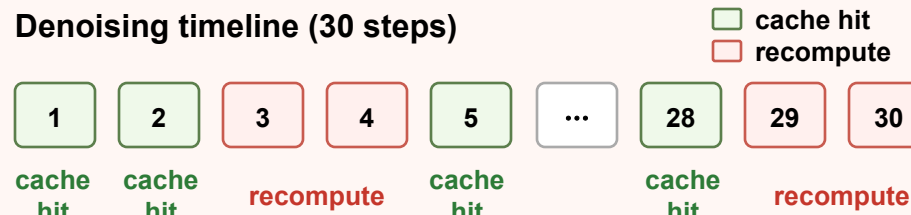


Challenges

1 Decision-Domain Mismatch in Cache+Quant

Denoising timeline (30 steps)

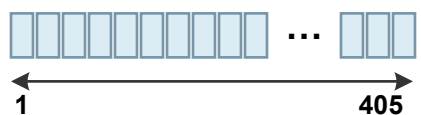


FP indicator $\neq$ INT8 execution relevance

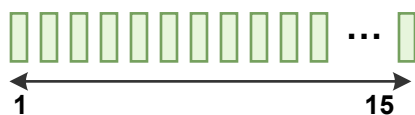


2 Matrix-Axis Mismatch in Factorized Attention

Spatial: 30 contexts $\times$ 405-token rows



Temporal: 810 contexts $\times$ 15-token rows



27 $\times$ more contexts

QKT: $[1, d] \times [d, L] \rightarrow$ irregularity on output axis

PV: $[1, L] \times [L, d] \rightarrow$ irregularity on reduction axis

Proposed Co-Design

1 QAC: Quantization-Aligned Caching

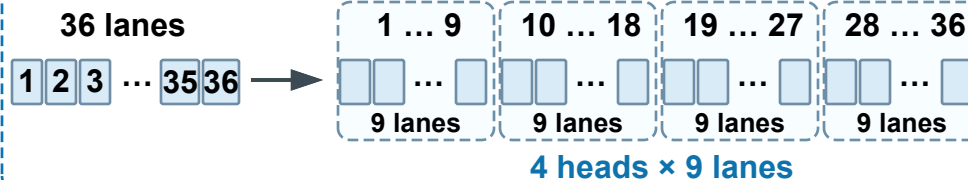
Quantized feature change estimation

Aligned cache criterion

Cache hit / Recompute decision

2 Dimension-Elastic PEA

① QK output-axis elasticity

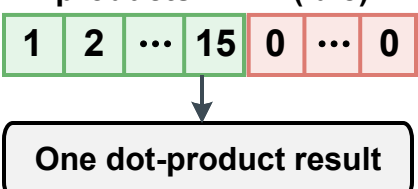


② PV reduction-axis elasticity

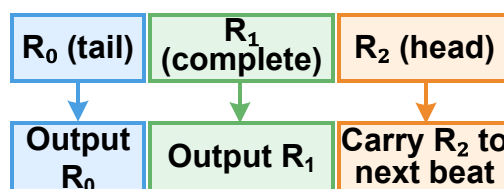
Baseline: padding for short rows (e.g., $T = 15$)

15 valid products, 21 padding (idle)

1 2 ... 15 0 ... 0



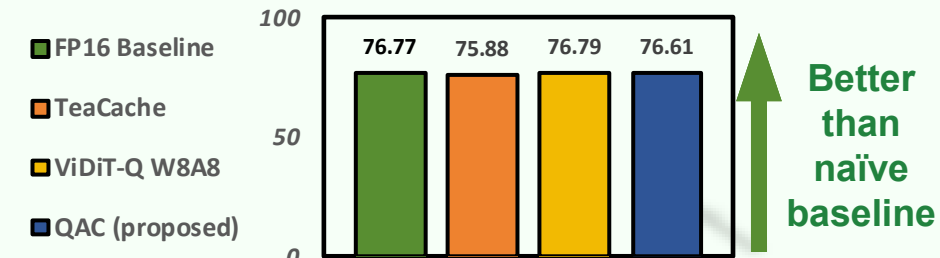
Proposed: boundary-aware segmented reduction



Benefits / Results

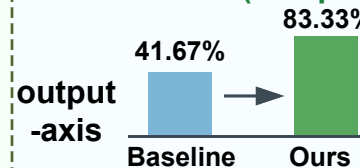
1 Algorithm Quality

VBBench (higher is better)

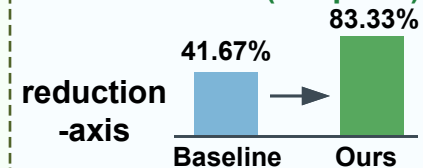


2 Hardware Efficiency

QK utilization (Temporal)



PV utilization (temporal)



3 System-Level Gains

Cache ratio:	$[BASE-CACHE-RATIO] \rightarrow [QAC-CACHE-RATIO]$
End-to-end speedup:	$[ATTN-SPEEDUP \times]$
Energy efficiency:	$[E2E-SPEEDUP \times]$
Area overhead:	$[ENERGY-EFF \times]$